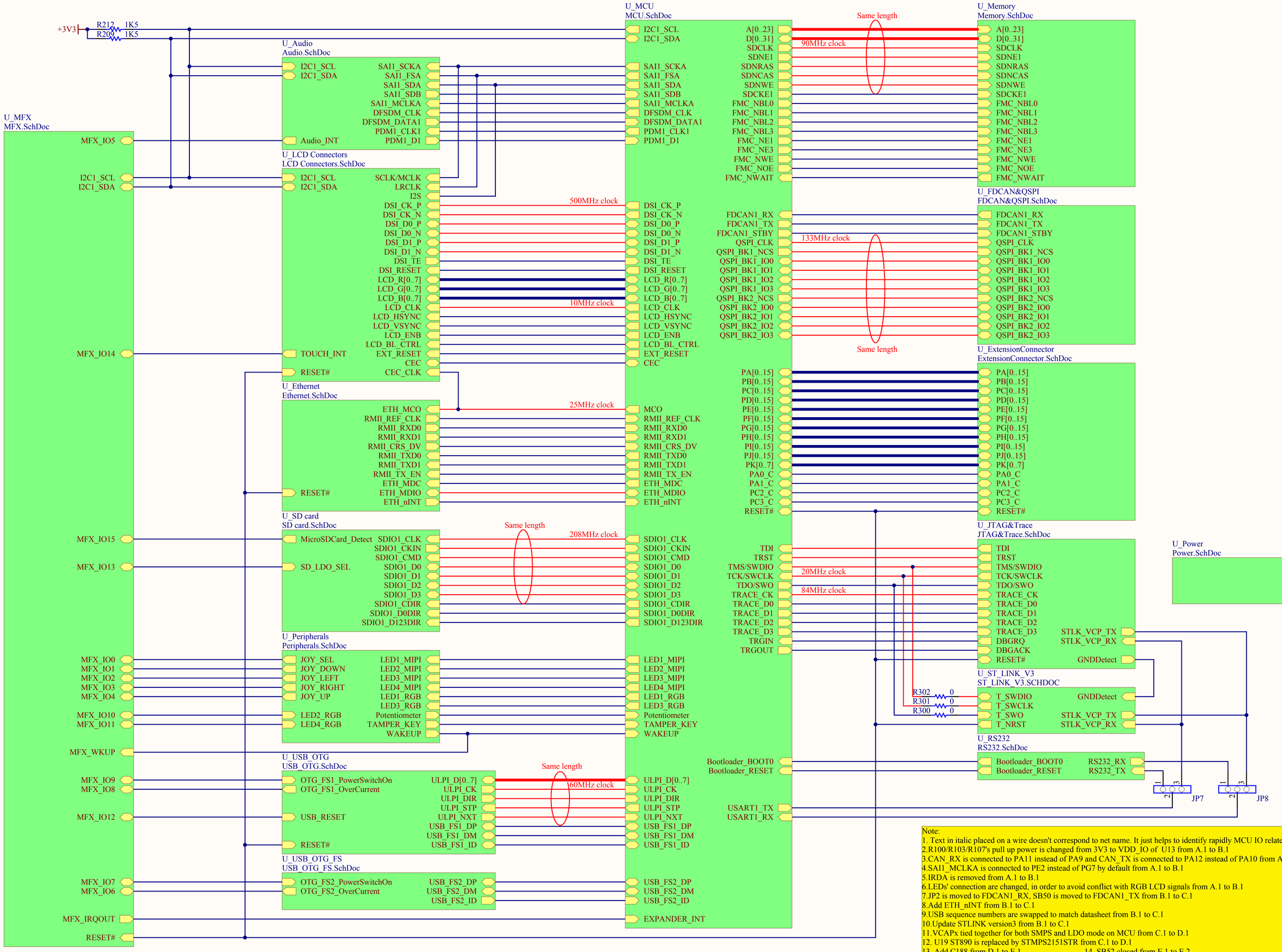
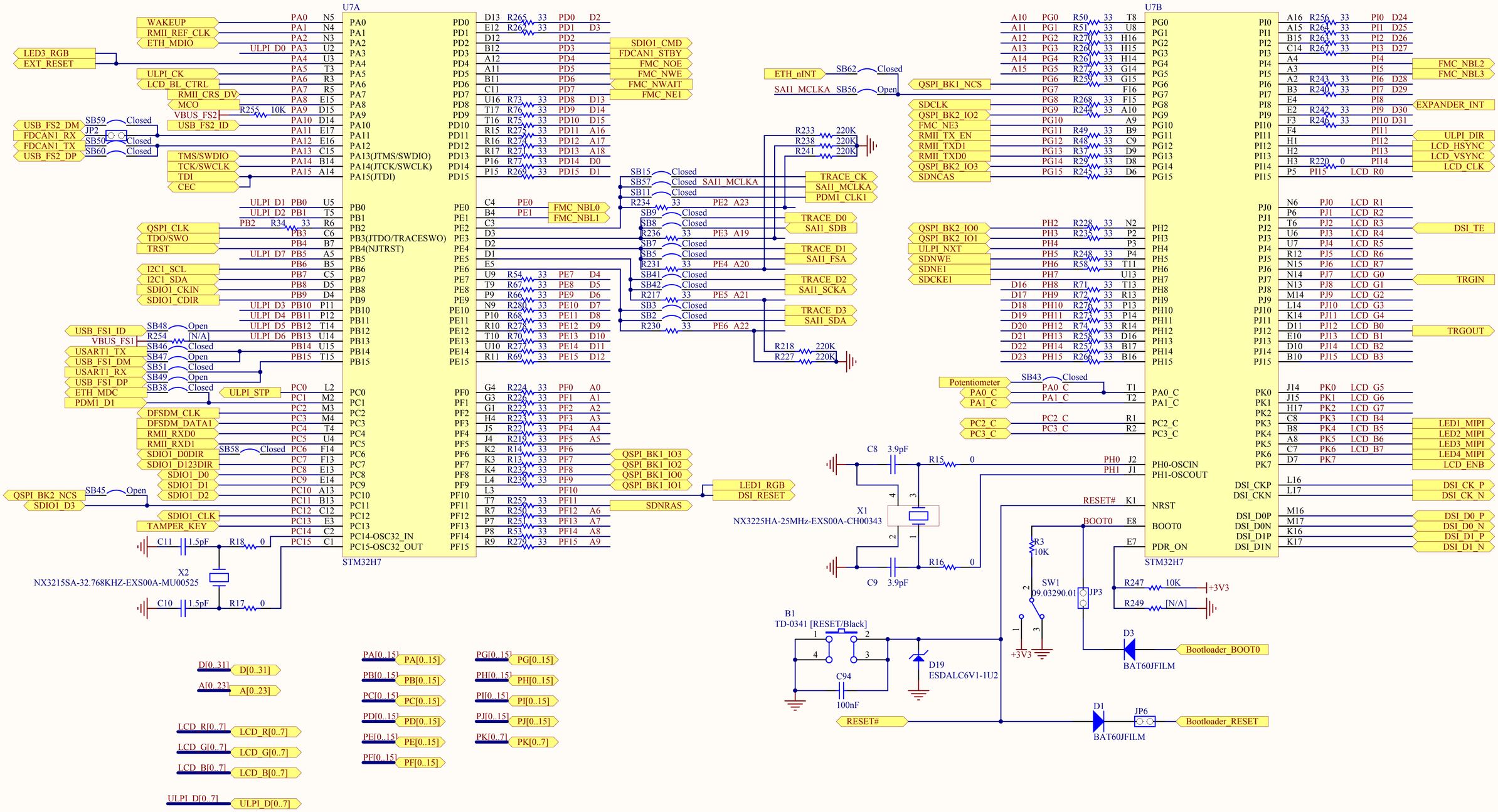


Note:
1. Text in italic placed on a wire doesn't correspond to net name. It just helps to identify rapidly MCU IO related to this wire.
2.R100/R103/R107's pull up power is changed from 3V3 to VDD_IO of U13 from A.1 to B.1
3.CAN_RX is connected to PA11 instead of PA9 and CAN_TX is connected to PA12 instead of PA10 from A.1 to B.1
4.SAI1_MCLKA is connected to PE2 instead of PG7 by default from A.1 to B.1
5.IRDA is removed from A.1 to B.1
6.LEDs' connection are changed, in order to avoid conflict with RGB LCD signals from A.1 to B.1
7.JP2 is moved to FDCAN1_RX, SB50 is moved to FDCAN1_TX from B.1 to C.1
8.Add ETH_nINT from B.1 to C.1
9.USB sequence numbers are swapped to match datasheet from B.1 to C.1
10.Update STLINK version3 from B.1 to C.1
11.VCAPx tied together for both SMPS and LDO mode on MCU from C.1 to D.1
12. U19 ST890 is replaced by STMPS2151STR from C.1 to D.1
13. Add C188 from D.1 to E.1 14. SB52 closed from E.1 to E.2

Title: **MB1246**		
Project: **STM32H7xxI-EVAL**		
Size: A3	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 1 of 17

life.augmented



Title: **MCU**			
Project: **STM32H7xxI-EVAL**			
Size: A3	Reference: MB1246	Revision: E.2	
Date: 2/13/2019		Sheet: 2 of 17	

life.augmented

Power configurations:

Mode	Mount	Removed
LDO mode STM32H7x3 Default setting	SB13,SB23,SB18	SB14,SB16,SB20 SB10,SB17,SB53 L1
SMPS mode STM32H7x7 Default setting	SB14,SB20,SB10 SB53 L1	SB13,SB17,SB23 SB16,SB18
SMPS+LDO mode	SB13,SB23,SB10 SB53, L1	SB14,SB16,SB17 SB18,SB20

U7C

STM32H7

VDD_MCU

VDD_SDC

L10 BEAD(FBMJ1608HM180NTR)

C12 10uF(GRM155R60G106ME44)

C188 220pF

L1 2.2uH(LQM2MPN2R2NG0)

SB53 C187 10uF

SMPS GND

Place near L1

C20 [N/A] SB16 Open

C28 4.7uF SB20

C15 4.7uF SB14

C18 2.2uF SB13

C30 2.2uF SB23

C22 2.2uF SB17 Open

SB10 SB18

VDD_MCU

Only for test

+5V SB21 Open

VDD_MCU

SB19 Closed

VDD_MCU

SB22 Closed

C29 1uF_X5R_0603

R2 0

VDDA

TP3 VDDA

R216 0

TP4 VREF

VREF+

+3V3

BT1 CR1220 holder

JP11

Removed for STM32H7x3

C135 100nF C26 2.2uF C27 2.2uF[N/A]

DSI_VDD12

Ceramic capacitor (Low ESR <600m ohm)
Only one 2.2uF required on board

D10 ZEN056V130A24LS

CN10 DC-10B

Z1 SMAJ5.0A-TR

C34 100nF

L3 BNX002-01

C36 220uF

E5V

JP10 Header 6X2

VBUS_HS VBUS_FS2 VBUS_FS1 D5V E5V U5V_ST_LINK

TP5 5V

+5V

R159 820

LD9 Green

+3V3 JP9 VDD_MCU

U20 LD1086D2M33

+5V C46 47uF

Vin Vout Tab GND

C51 10uF

+3V3 C52 100nF

TP7 +3V3

TP11 Ground TP1 Ground TP6 Ground TP2 Ground

U17 ST1L05BPUR

+5V

+3V3 R143 1K

VI EN PG VO ADJ GND GND

C54 [N/A] C50 4.7uF

TP9 2V9_SD

+2V9_SD

R1 R134 22K[1%]

C48 [N/A]

R2 R136 16K[1%]

C47 10uF [Ceramic]

Power Supply 2.9V
Vout=1.22*(1+R1/R2)

U21 LD1117S18TR

+3V3 Vin Vout Tab

VDD_SDC SB36 Open

+1V8_LDO SB37 Closed

+1V8

TP12 1V8

C59 10uF

C124 C115 C123 C130 C129 C131 C132 C134 C118 C128 100nF

C21 4.7uF

C125 C119 C127 C122 C117 C133 C126 C121 C114 C120 100nF

VREF+ C106 1uF C110 100nF

VDDA C107 1uF C109 100nF

Title: **Power**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 3 of 17

life.augmented

D[0..31]
A[0..23]

U16
IS61WV102416BLL-10MLI
SRAM

U12
MT28EW128ABA1LPC-0SIT
Nor Flash

U15
IS42S32800G-6BLI
SDRAM

+3V3
R129 10K
FMC_NE3 PG10 R128 0
FMC_NBL0 PE0
FMC_NBL1 PE1
R87 10K
R89 10K
R28 10K
FMC_NE1 PD7 R88 0
FMC_NWE PD5 R95 0
FMC_NOE PD4
FMC_NWAIT PD6
R86 10K
R96 10K
C33 100nF
Default setting: Open
JP13

SDNWE PH5
SDNCAS PG15
SDNRAS PF11
SDNE1 PH6
SDCKE1 PH7 R282 0
SDCLK PG8
FMC_NBL3 PI5
FMC_NBL2 PI4
FMC_NBL1
FMC_NBL0

C155 C146 100nF 100nF
Place close SRAM

C138 C136 C137 100nF 100nF 100nF
Place close Nor Flash

C159 C158 C157 C143 C145 C161 C160 C156 C142 C144 C150 C152 C151 C149 100nF
Place close SDRAM

Title: **SRAM & FLASH & SDRAM**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 4 of 17

life.augmented

Default I2C Address:0011010

U22

WM8994ECS/R

MS-23D18

MP34DT05TR-A

U26 *RIGHT*

U27 *LEFT*

MP34DT05TR-A

JP12

To MFX

time slot0 of SAI

time slot1 of SAI

D17 ESDA6V1BC6

D18 ESDA6V1BC6

PJ3028B-3

PJ3028B-3

Coupon connectors

RIGHT

CN26

Coupon Connectors

LEFT

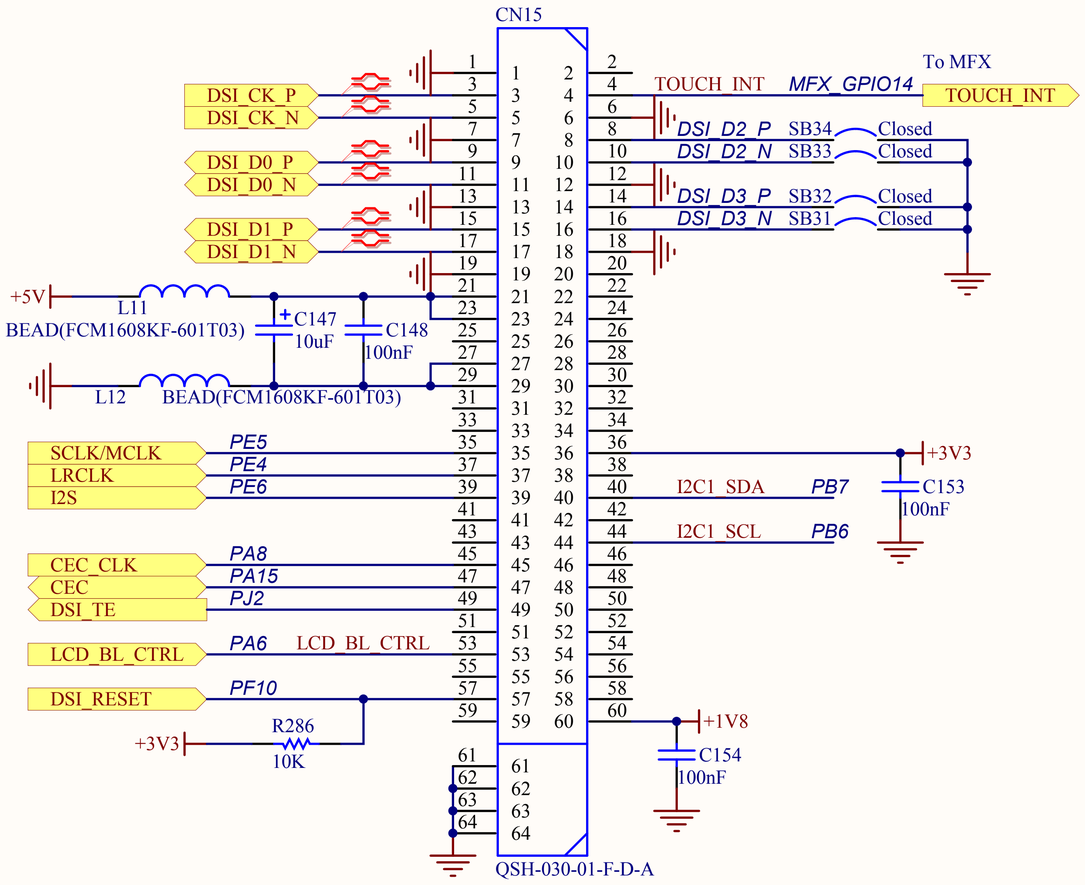
CN25

Coupon Connectors

Title: **Audio**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 5 of 17

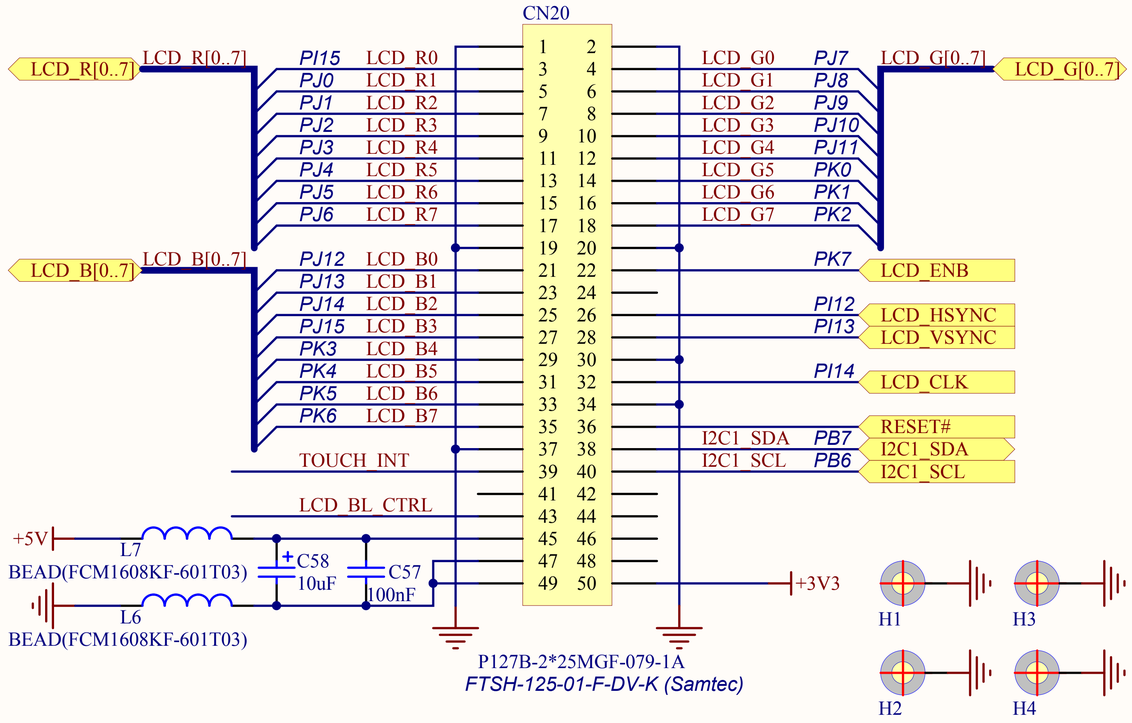
life.augmented

DSI LCD

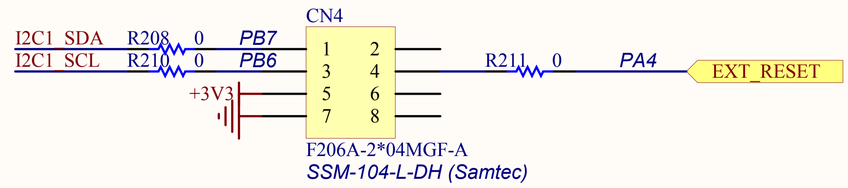


Removed for STM32H7x3

RGB LCD

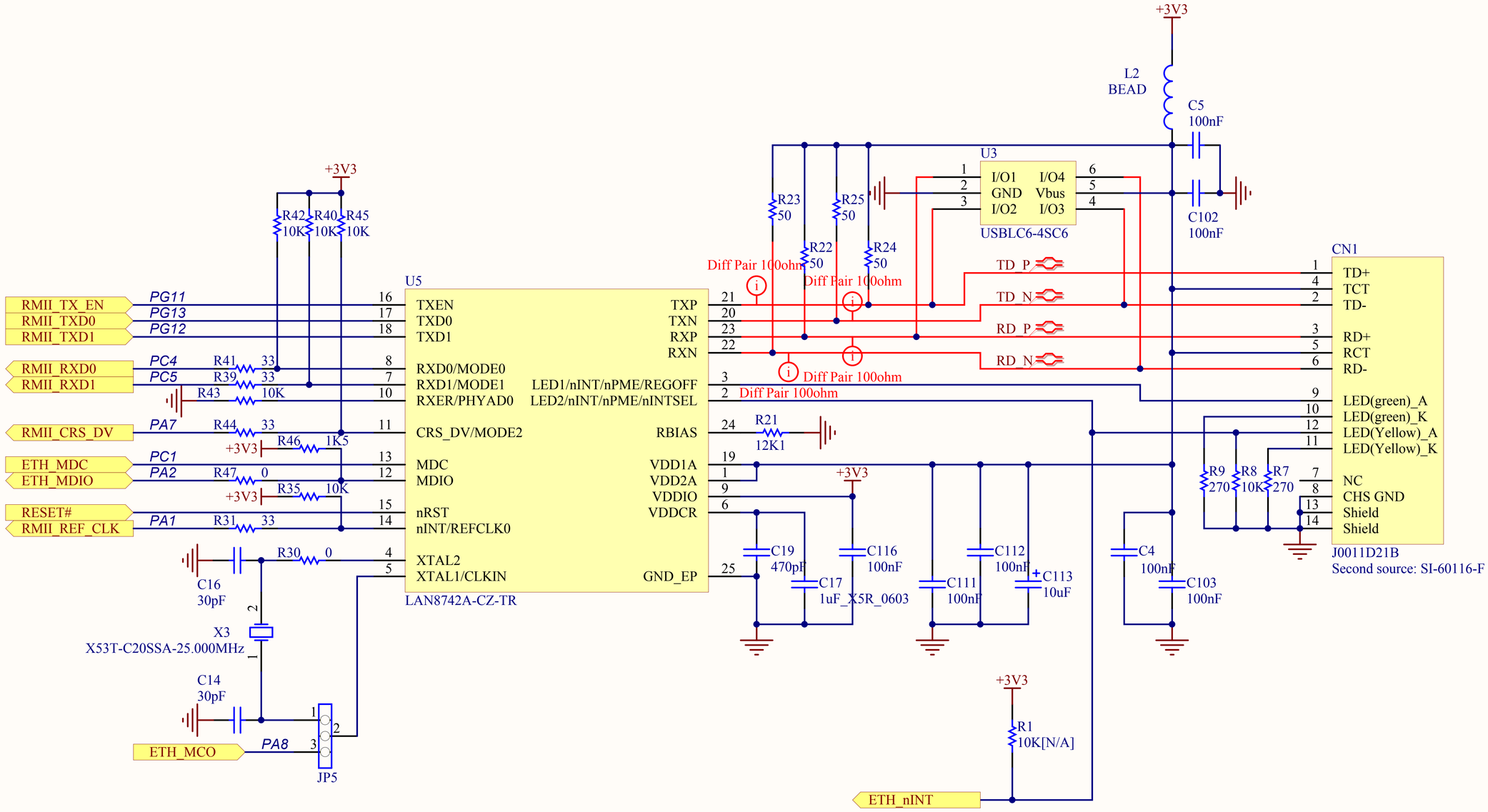


I2C Connector



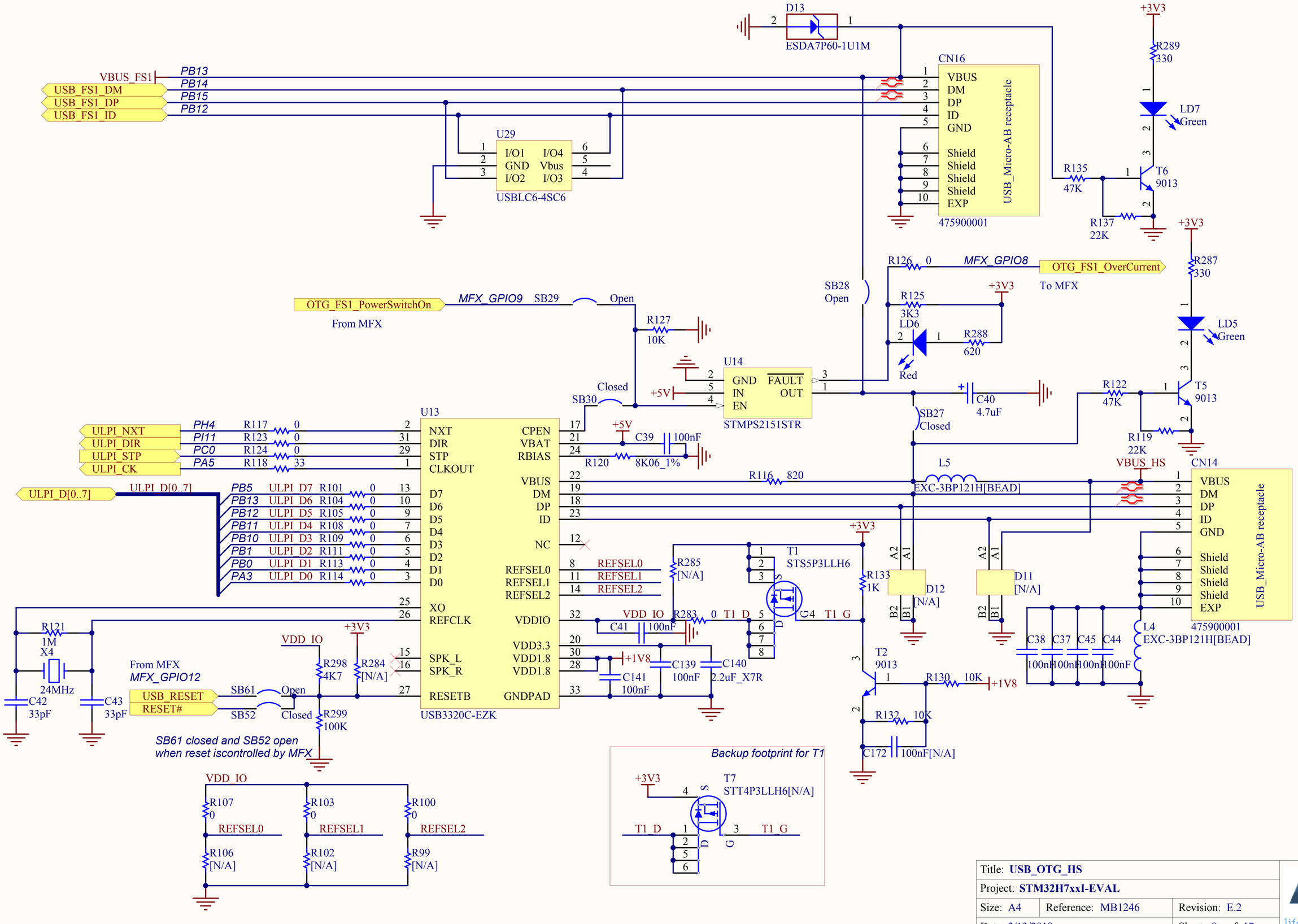
Title: **LCD Connectors**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 6 of 17

life.augmented



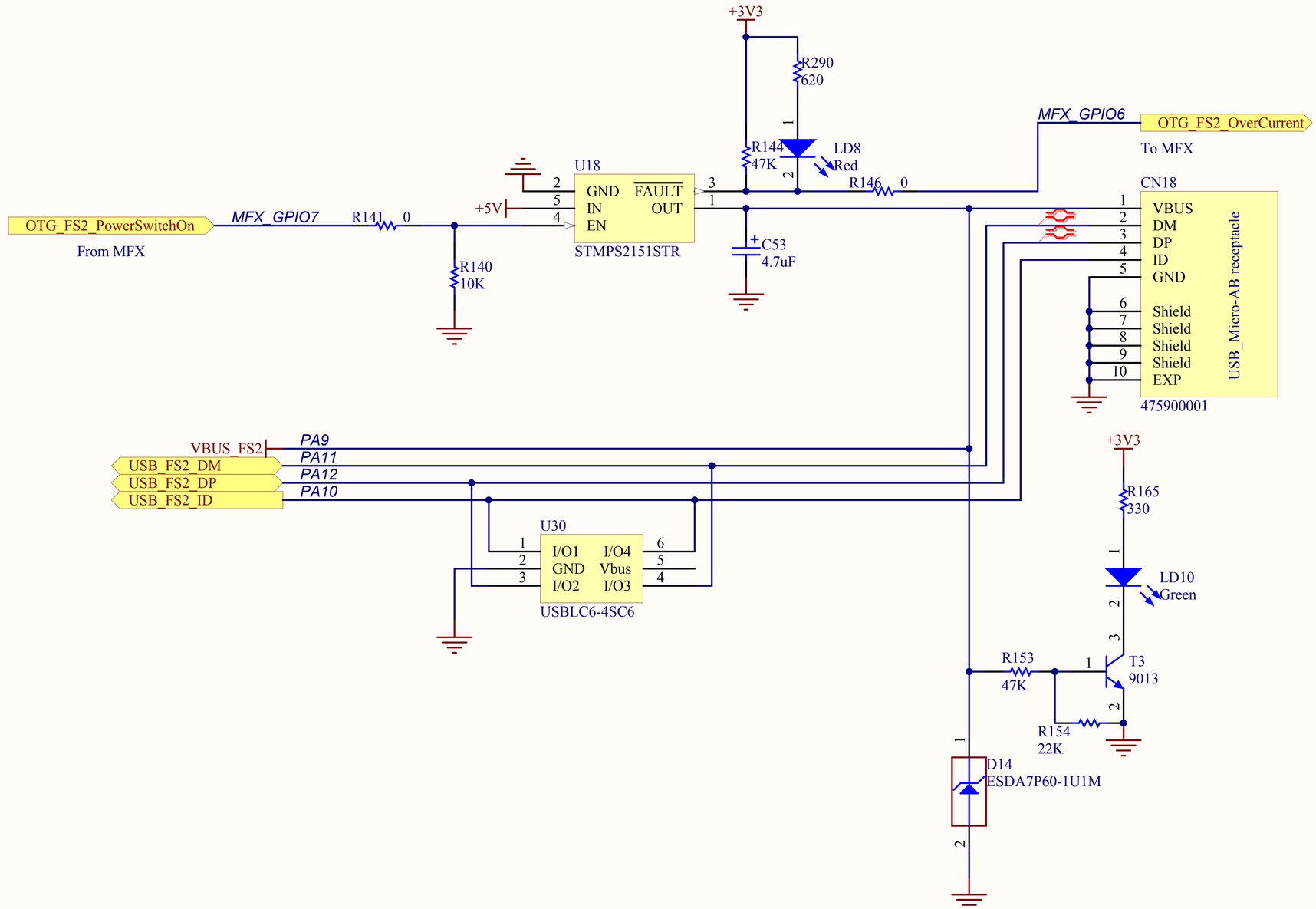
Title: **Ethernet**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 7 of 17

life.augmented



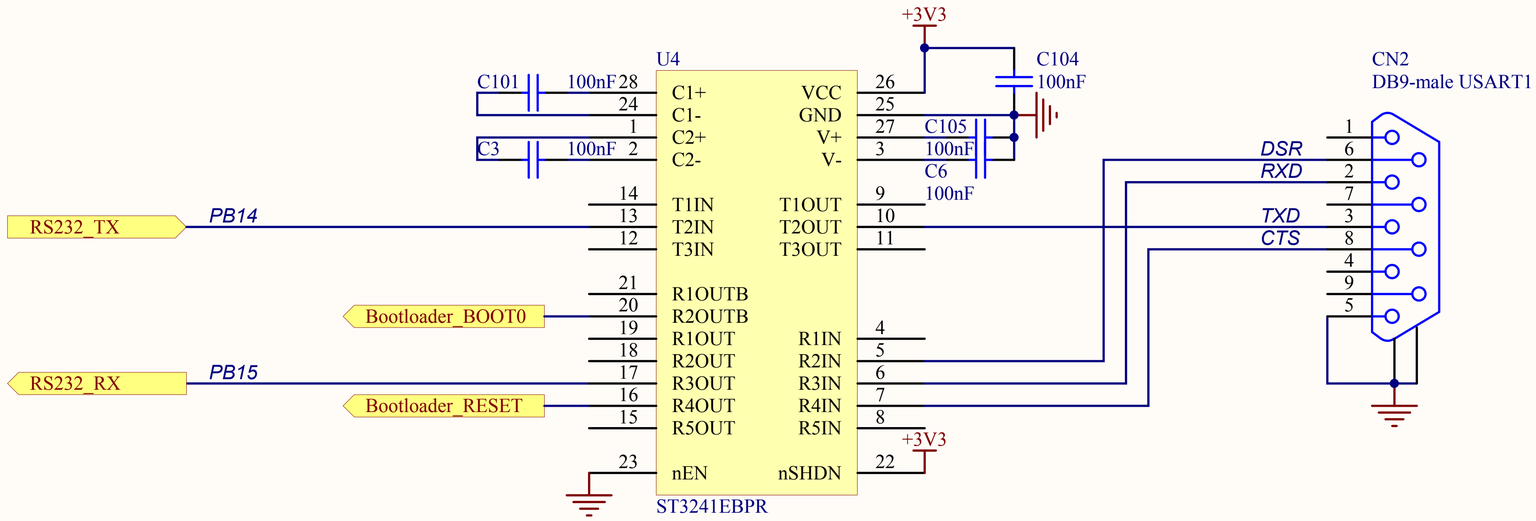
Title: **USB_OTG_HS**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 8 of 17

life.augmented



Title: **USB_OTG_FS**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 9 of 17

life.augmented

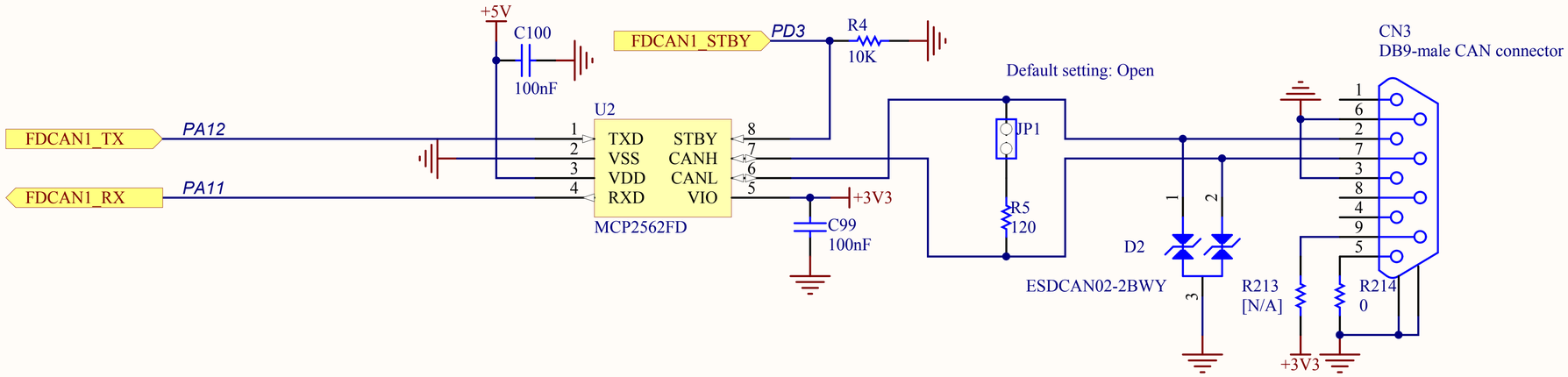


RS232

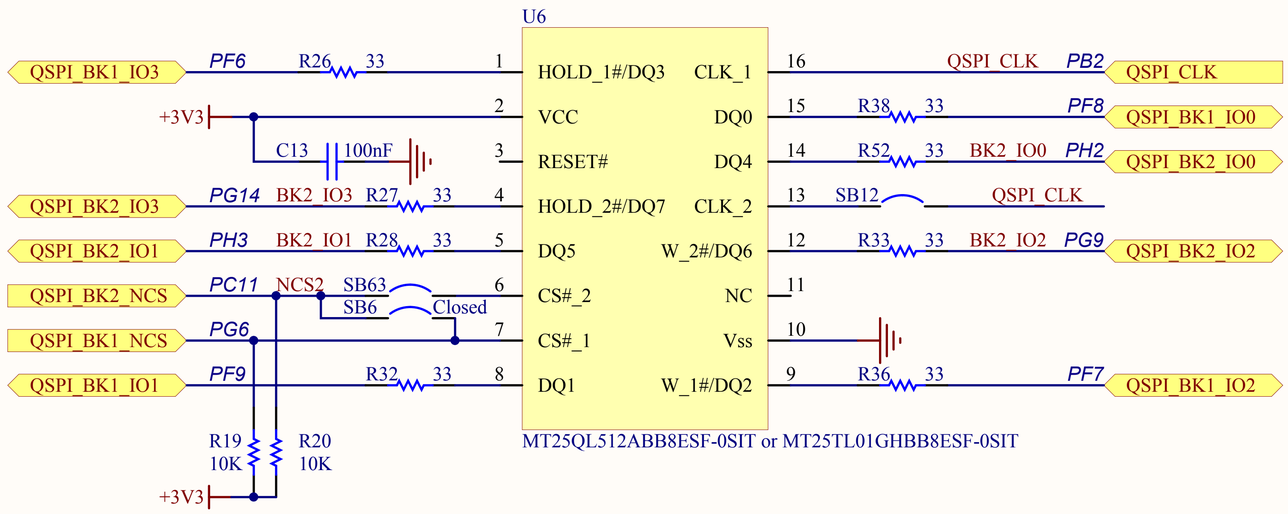
Title: **RS232**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 10 of 17

life.augmented

FDCAN



Twin Quad SPI Flash



Removed when using One Twin Quad SPI Flash

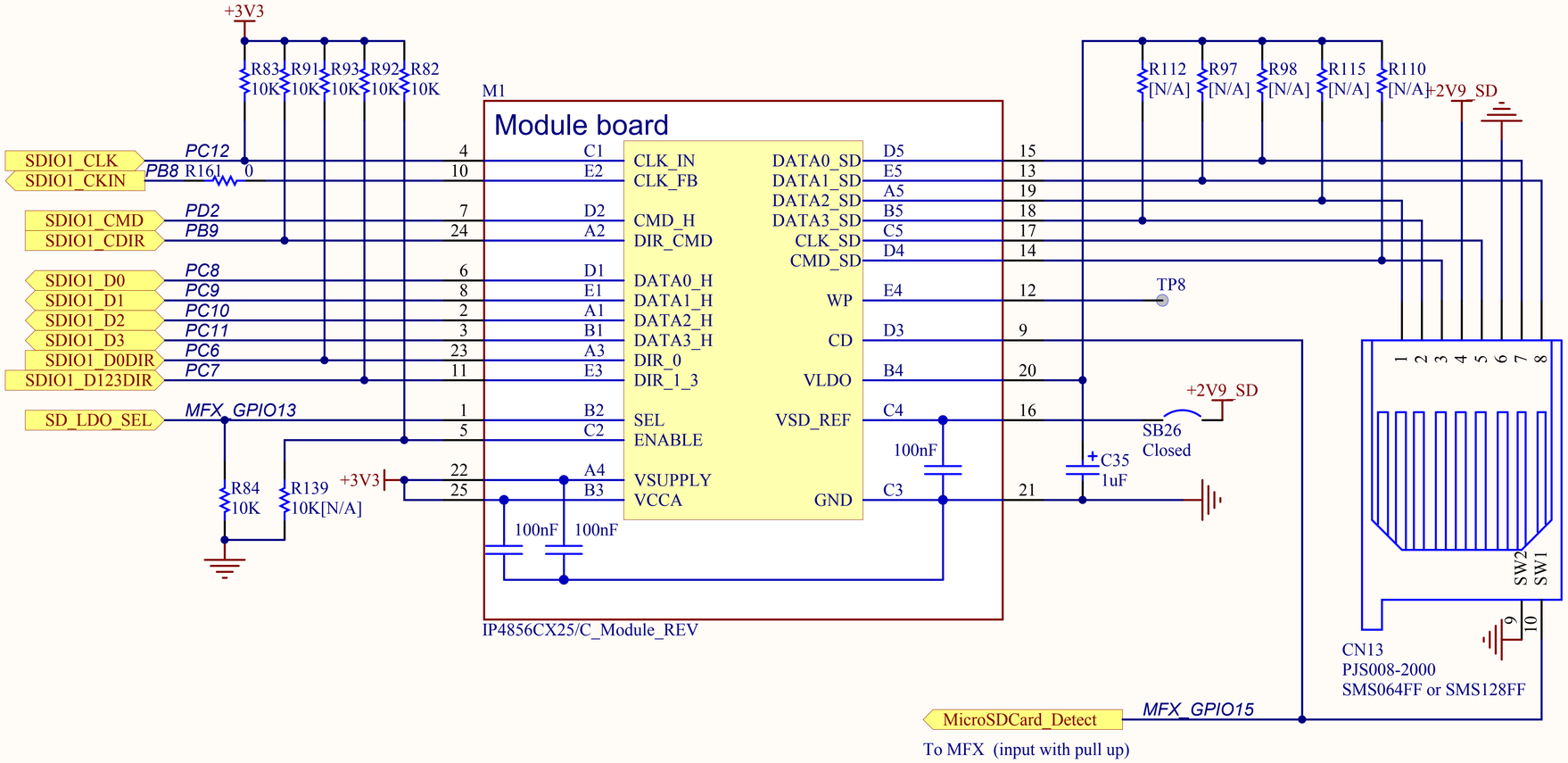
Twin Quad SPI Flash configurations:

Two Quad SPI Flash	U6:MT25QL512ABB8ESF-0SIT	U28:MT25QL512ABB8ESF-0SIT	R225/R229/R237/R253, SB44, C108 ON	R27/R28/R33/R52, SB12/SB63 OFF
One Twin Quad SPI Flash	U6:MT25TL01GHBB8ESF-0SIT	U28:NA	R27/R28/R33/R52, SB12/SB63 ON	R225/R229/R237/R253, SB44, C108 OFF

Title: **FDCAN & QSPI**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 11 of 17

life.augmented

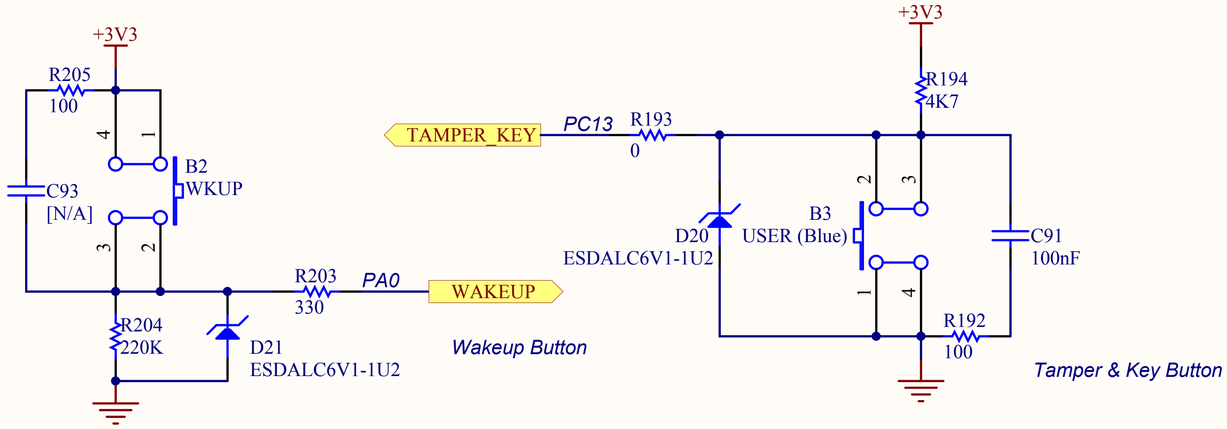
MICRO SD 3.0 Card



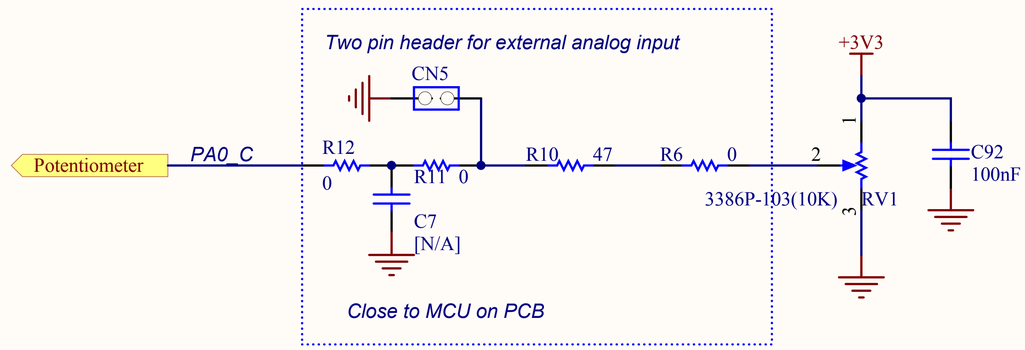
Title: **MICRO SD 3.0 Card**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 12 of 17

life.augmented

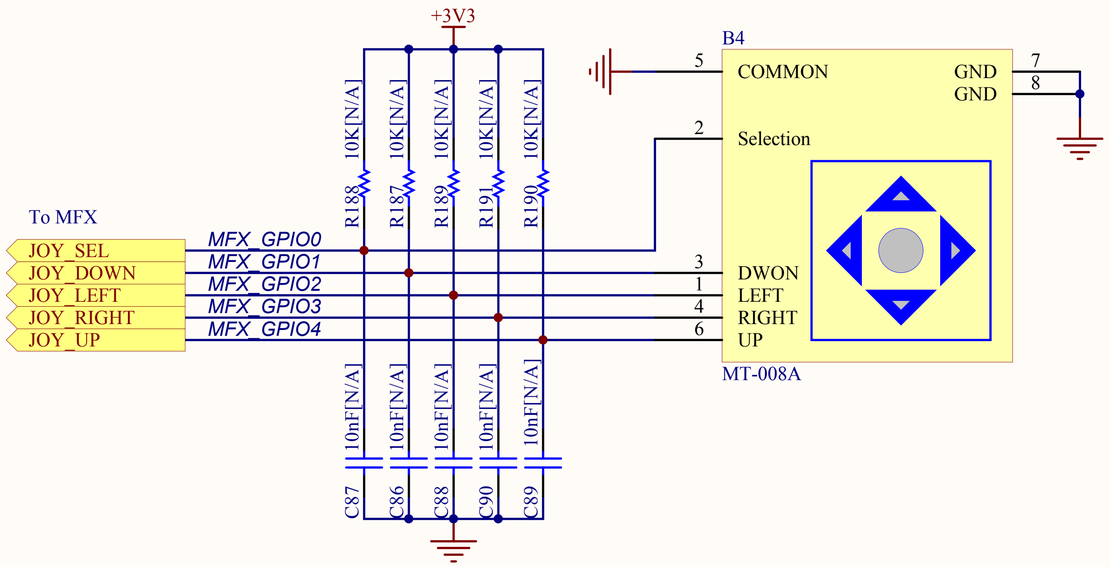
Buttons



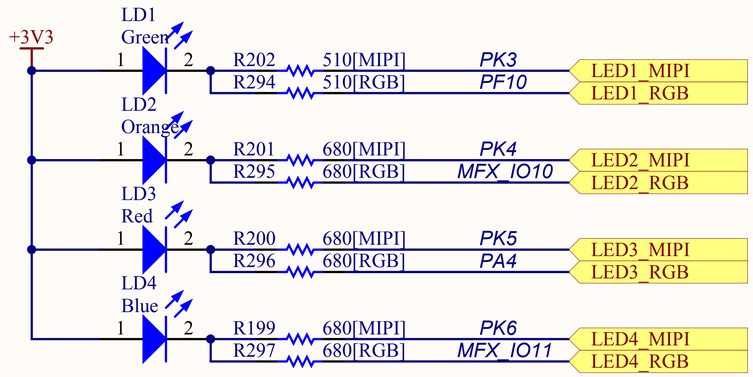
Potentiometer



Joystick

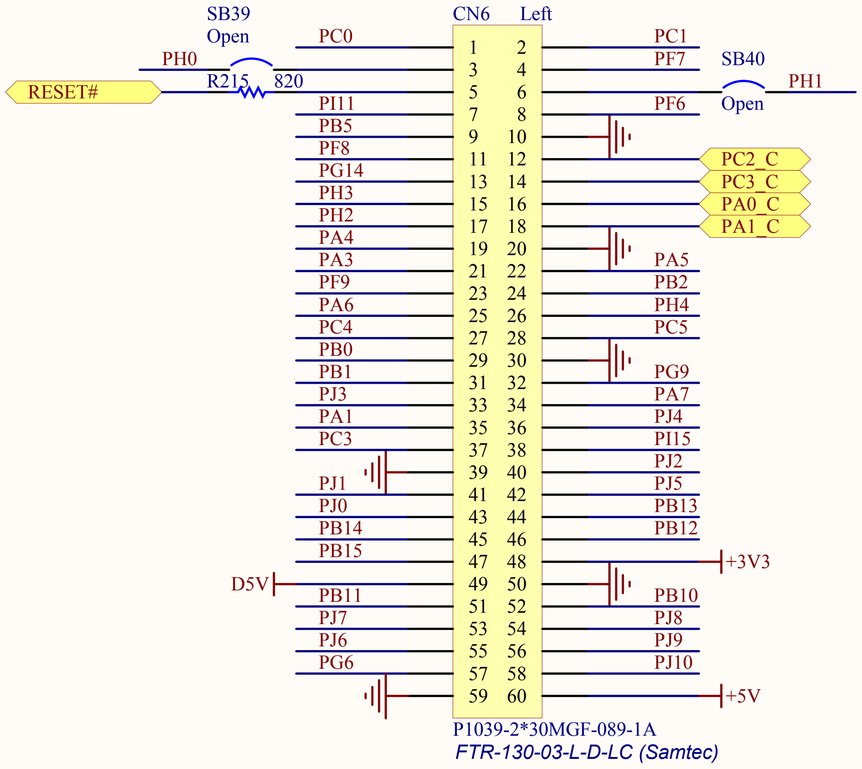


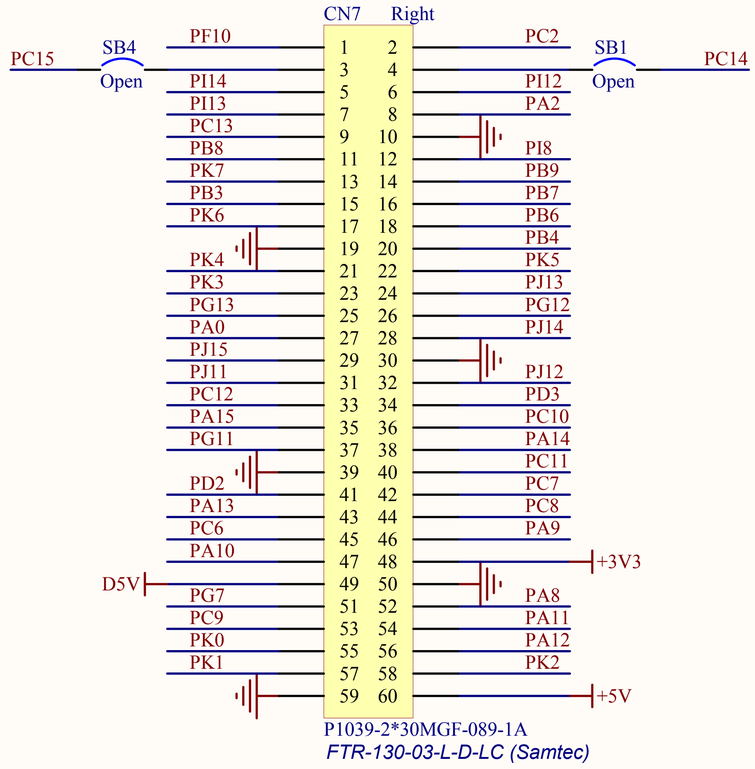
LEDs

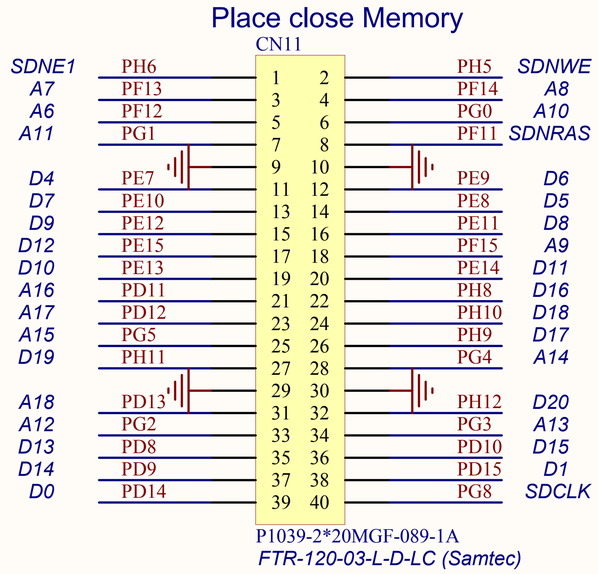


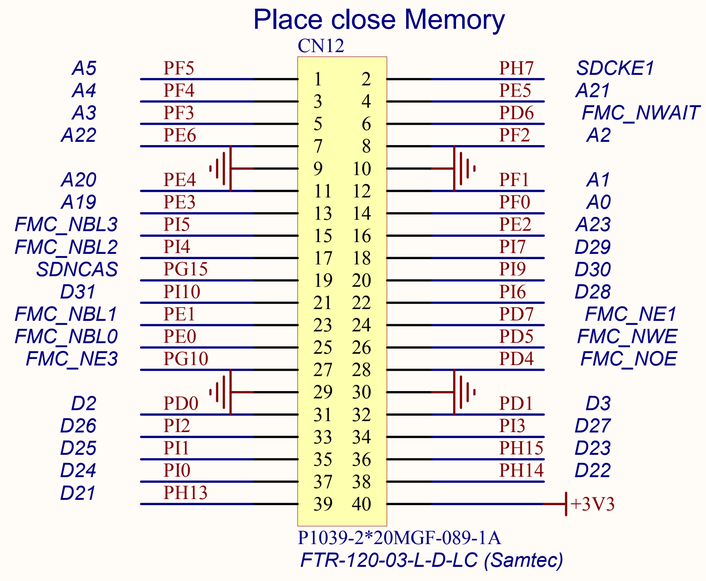
Title: **Peripherals**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 13 of 17

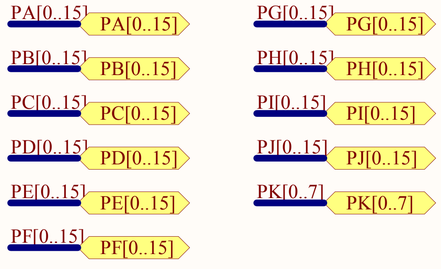
life.augmented











Title: **Extension Connectors**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 14 of 17

life.augmented

5 Volts ST-LINK

CN21
Fitted: NO
Silkscreen = Reserved
4
3
2
1
Header 4x1
STLK_SWDIO
STLK_SWCLK
+3V3_ST_LINK
USB_DEV_HS_N
USB_DEV_HS_P

U23A
STLK_SWDIO/PA13 A15
STLK_SWCLK/PA14 A14
USB_DEV_HS_N/PB14 R14
USB_DEV_HS_P/PB15 R15
OSC_IN/PH0 G1
OSC_OUT/PH1 H1
NRST J1
BOOT0 D6
PDR_ON C6
V12_OTGPHY J14
REXT_OTGPHY J15
T_SW_DIR/PA7 R3
T_SWDIO_IN/PH7 N12
T_SWDIO_OUT/PF9 L2 T_SWDIO
T_SWCLK/PH6 M11 T_SWCLK
T_SWO/PD2 D12 T_SWO
STLK_VCP_RX/PG9 C10 STLK_VCP_RX
STLK_VCP_TX/PG14 A7 STLK_VCP_TX
GNDDetect/PG5 K13 GNDDetect
T_VCC AIN/PA0 N3
T_PWR_EXT/PB1 R4 T_PWR_EXT
T_PWR_EN/PB0 R5 T_PWR_EN
STLK_LED/PA10 D15 STLK_LED
STLK_MCO/PA8 F15
T_NRST/PA5 P4 T_NRST
HW_TYP_0/PI4 D4
HW_TYP_1/PI5 C4
ST-LINKV3E_SWD

+3V3_ST_LINK
X5
VDD OUT EN GND
NZ2520SH-25.00M
R175 10K
C1 100nF
R179 100
JP4
C164 100nF
Fitted: NO
+3V3_ST_LINK

T_NRST
T_SWO
T_SWDIO
T_SWCLK
JP14
Fitted: NO

C78 2.2uF
R198 3K

STLK_VCP_RX
STLK_VCP_TX
R164 4K7 T_VCC
R167 4K7
+3V3
STLK_MCO
C81 20pF
Fitted: NO

VUSB_ST_LINK
R195 1K
RED 2V
LD11
R142 100K
U19
IN OUT EN FAULT GND
STMPS2151
U5V_ST_LINK
R150 10K
C56 1uF
C49 100nF

+5V
D16 BAT60JFILM
+3V3_ST_LINK
SB35 Open
R292 2K7
R293 4K7

LD12
Red
Green
R196 330
R197 330
+3V3_ST_LINK
LD_BICOLOR_CMS

3.3 Volts ST-LINK

VBUS_HS
VBUS_FS2
VBUS_FS1
E5V
D5V
VUSB_ST_LINK
D4 BAT60JFILM
D5 BAT60JFILM
D6 BAT60JFILM
D8 BAT60JFILM
D7 BAT60JFILM
D9 BAT60JFILM

U9 LD3985M33R
Vin Vout INH GND BYPASS
+3V3_ST_LINK
C32 1uF
C31 100nF
C25 10nF
C23 1uF
C24 100nF

+3V3_ST_LINK
C184 100nF
C177 100nF
C165 100nF
C178 100nF
C2 100nF
C180 100nF
C181 100nF
C162 100nF
C174 100nF
C183 100nF
C170 100nF
C166 100nF
C179 100nF
C98 100nF
C173 100nF
C176 100nF
C97 4.7uF
C175 1uF
C182 100nF

R177 10K
C186 2.2uF
C185 2.2uF
Ceramic capacitor (Low ESR)

U23B
ST-LINKV3E_SWD

Impedance Constraint [Min = 85.00 Max = 95.00]
Matched Net Lengths [Tolerance = 0.2mm]

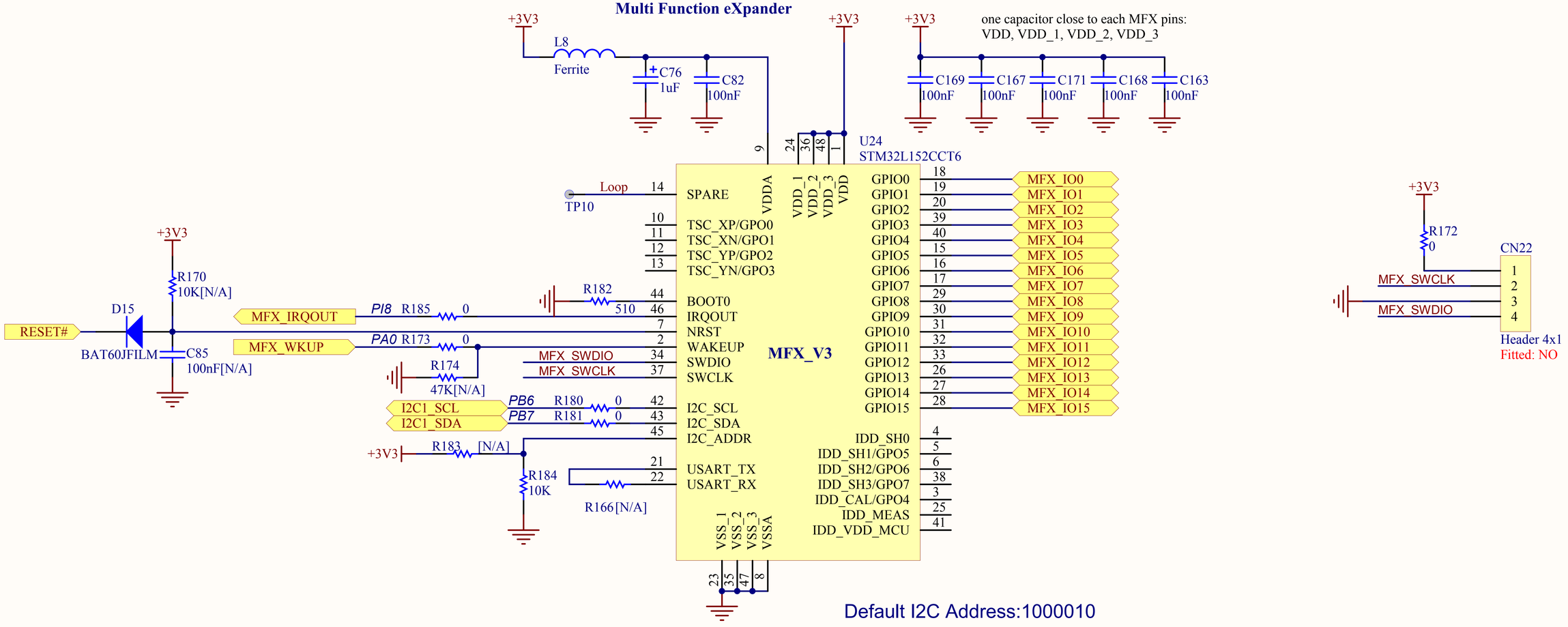
CN23
USB_Micro-B receptacle
VBUS DM DP ID GND Shield Shield Shield Shield EXP EXP
USB_uB_105017-0001
VUSB_ST_LINK
C67 100nF
D22
Fitted: NO
ESDA7P60-1U1M
Closed to Pin1 of Mirco-B

USB_DEV_HS_CN_N
USB_DEV_HS_CN_P
USB
U25
NC GND D-in D-out D+in D+out
ECMF02-2AMX6
USB_DEV_HS_N
USB_DEV_HS_P

Title: **ST-LINK/V3E**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 15 of 17

life.augmented

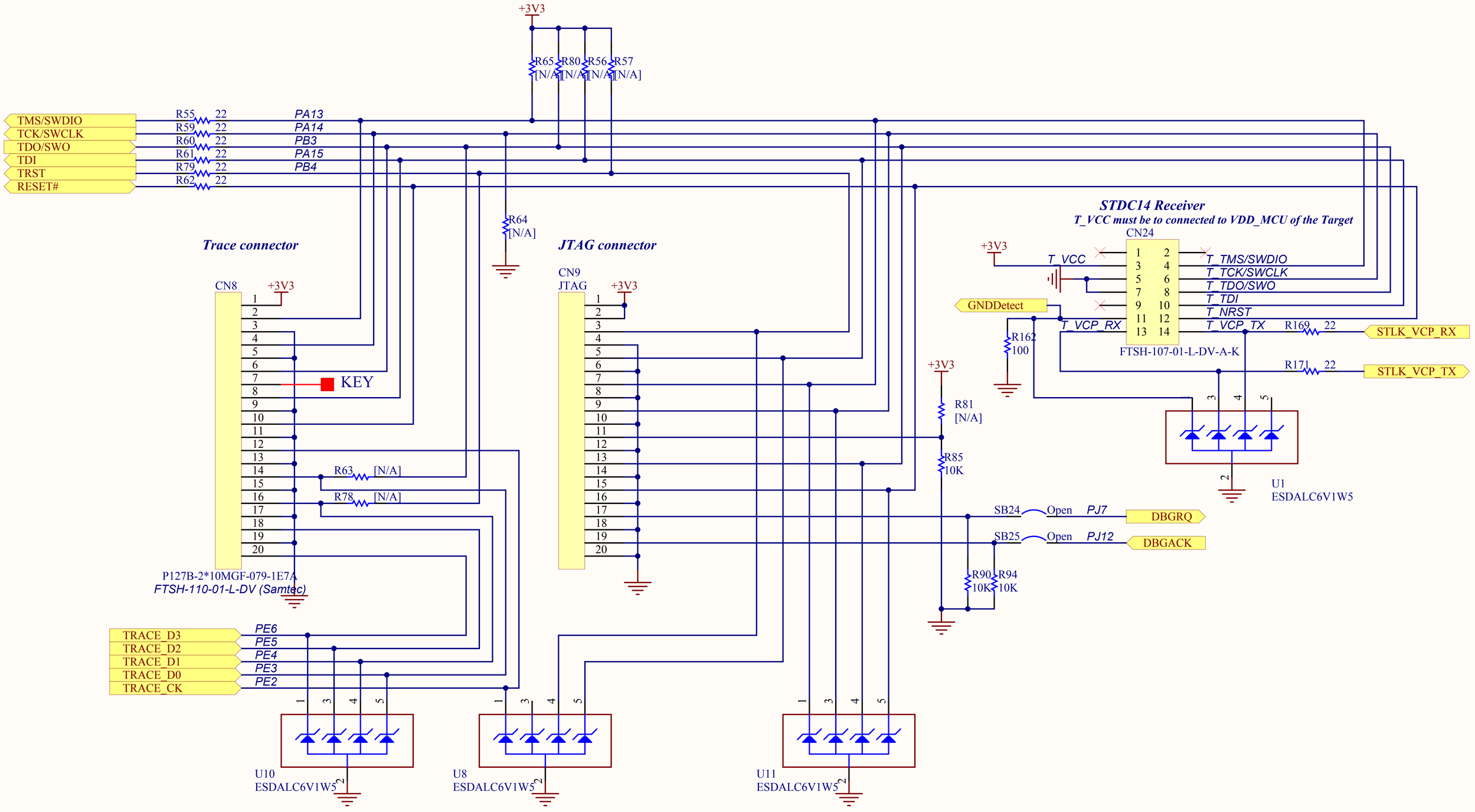
MFX
Multi Function eXpander



Default I2C Address:1000010

Title: **MFX**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 16 of 17

life.augmented



Title: **JTAG & Trace**		
Project: **STM32H7xxI-EVAL**		
Size: A4	Reference: MB1246	Revision: E.2
Date: 2/13/2019		Sheet: 17 of 17

life.augmented